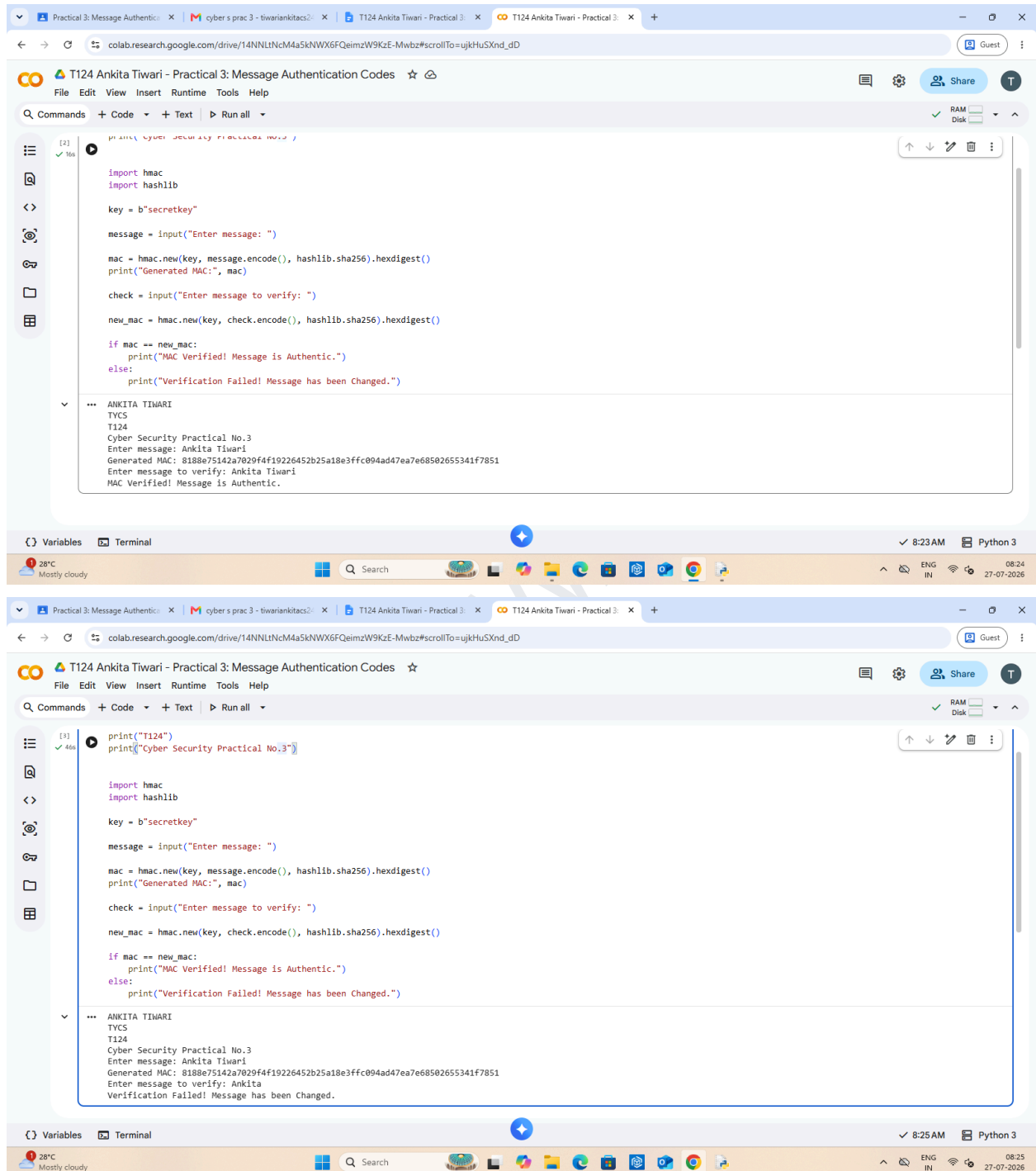


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AIM: Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

OUTPUT:



The image displays two screenshots of a Google Colab notebook titled "T124 Ankita Tiwari - Practical 3: Message Authentication Codes". The notebook is written in Python and demonstrates the generation and verification of Message Authentication Codes (MACs) using the HMAC library.

First Screenshot: The code is executed, and the output shows the MAC generation process. The user enters the message "Ankita Tiwari", and the generated MAC is "8188e75142a7029f4f19226452b25a18e3ffc094ad47ea7e68502655341f7851". The verification step confirms that the MAC is valid, printing "MAC Verified! Message is Authentic."

```
import hmac
import hashlib

key = b"secretkey"

message = input("Enter message: ")

mac = hmac.new(key, message.encode(), hashlib.sha256).hexdigest()
print("Generated MAC:", mac)

check = input("Enter message to verify: ")

new_mac = hmac.new(key, check.encode(), hashlib.sha256).hexdigest()

if mac == new_mac:
    print("MAC Verified! Message is Authentic.")
else:
    print("Verification Failed! Message has been Changed.")
```

Second Screenshot: The code is executed again, but this time the user enters a different message, "Ankita". The generated MAC is "8188e75142a7029f4f19226452b25a18e3ffc094ad47ea7e68502655341f7851". The verification step fails, printing "Verification Failed! Message has been Changed."

```
print("T124")
print("Cyber Security Practical No.3")

import hmac
import hashlib

key = b"secretkey"

message = input("Enter message: ")

mac = hmac.new(key, message.encode(), hashlib.sha256).hexdigest()
print("Generated MAC:", mac)

check = input("Enter message to verify: ")

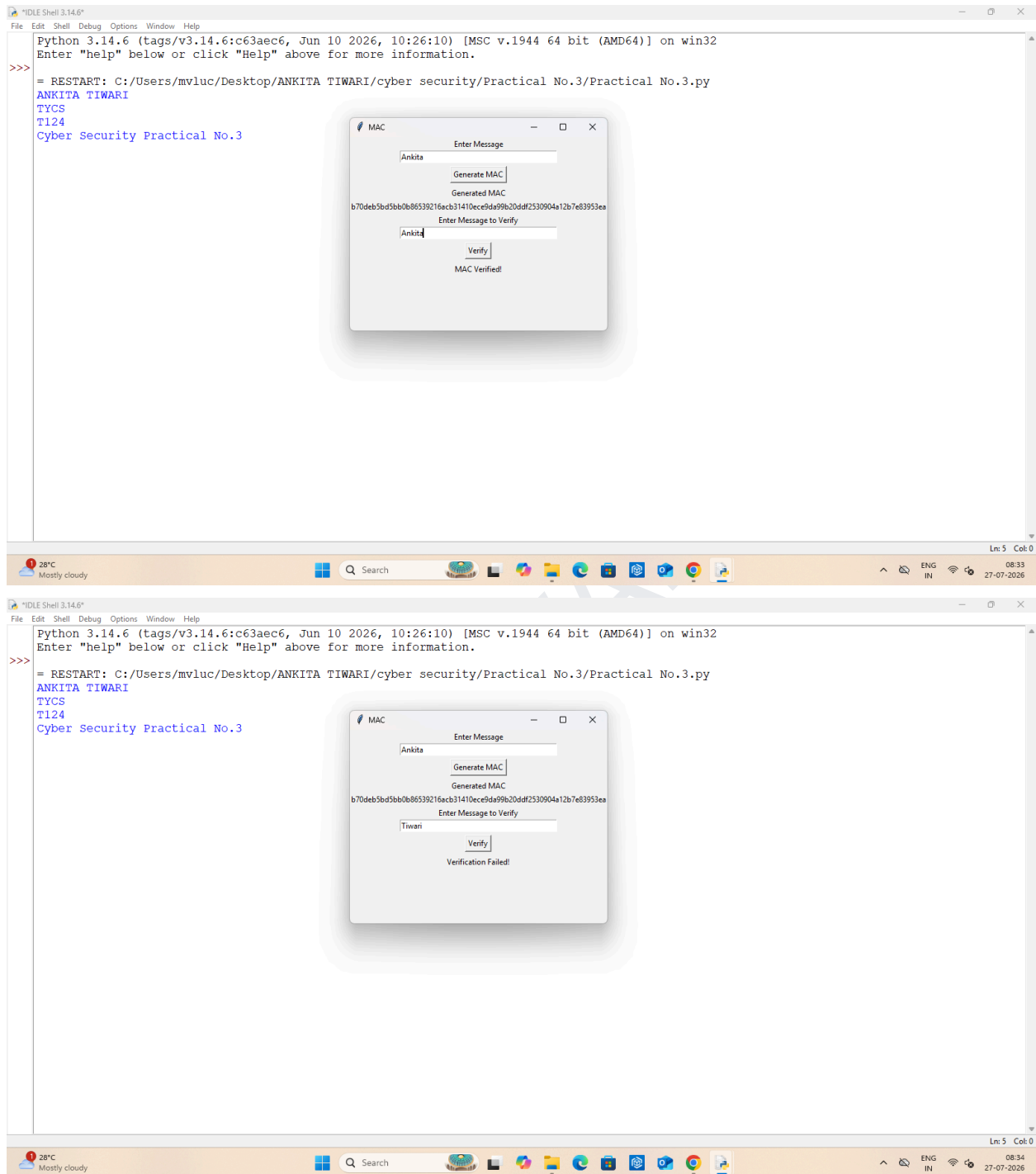
new_mac = hmac.new(key, check.encode(), hashlib.sha256).hexdigest()

if mac == new_mac:
    print("MAC Verified! Message is Authentic.")
else:
    print("Verification Failed! Message has been Changed.")
```

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GUI:



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